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## -----
# CITY LIBRARY MANAGEMENT SYSTEM
# -----

books = []
members = []
borrow_log = []

# -----
# 1. ADD BOOK
# -----
def add_book():
    book_id = input("Enter Book ID: ")
    title = input("Enter Title: ")
    author = input("Enter Author: ")
    genre = input("Enter Genre: ")

    book = {
        "BookID": book_id,
        "Title": title,
        "Author": author,
        "Genre": genre,
        "Availability": "Available"
    }
    books.append(book)
    print("Book added successfully!\n")

# -----
# 2. ADD MEMBER
# -----
def add_member():
    member_id = input("Enter Member ID: ")
    name = input("Enter Name: ")
    age = int(input("Enter Age: "))
    contact = input("Enter Contact Info: ")

    member = {
        "MemberID": member_id,
        "Name": name,
        "Age": age,
        "Contact": contact,
        "BorrowedBooks": []
    }
    members.append(member)
    print("Member added successfully!\n")

# -----

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# 3. ISSUE BOOK
# -----
def issue_book():
    member_id = input("Enter Member ID: ")
    book_id = input("Enter Book ID to issue: ")

    # find member
    member = None
    for m in members:
        if m["MemberID"] == member_id:
            member = m
            break

    if not member:
        print("Member not found!\n")
        return

    # find book
    book = None
    for b in books:
        if b["BookID"] == book_id:
            book = b
            break

    if not book:
        print("Book not found!\n")
        return

    if book["Availability"] == "Issued":
        print("Book already issued!\n")
        return

    # issue book
    book["Availability"] = "Issued"
    member["BorrowedBooks"].append(book_id)
    borrow_log.append({"MemberID": member_id, "BookID": book_id,
"Genre": book["Genre"]})

    print("Book issued successfully!\n")

# -----
# 4. RETURN BOOK
# -----
def return_book():
    member_id = input("Enter Member ID: ")
    book_id = input("Enter Book ID to return: ")

    # find member
    member = None

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for m in members:
    if m["MemberID"] == member_id:
        member = m
        break

if not member:
    print("Member not found!\n")
    return

if book_id not in member["BorrowedBooks"]:
    print("This member did not borrow the book.\n")
    return

# find book
for b in books:
    if b["BookID"] == book_id:
        b["Availability"] = "Available"
        break

member["BorrowedBooks"].remove(book_id)
print("Book returned successfully!\n")

# -----
# 5. REPORT – Available Books by Genre
# -----
def available_books_by_genre():
    genre = input("Enter genre: ")
    print(f"\nAvailable books in {genre}:")

    found = False
    for b in books:
        if b["Genre"].lower() == genre.lower() and b["Availability"]
== "Available":
            print(f"- {b['Title']} by {b['Author']}")
            found = True

    if not found:
        print("No available books found.\n")
    print()

# -----
# 6. REPORT – Members who borrowed books
# -----
def members_with_borrowed_books():
    print("\nMembers who borrowed books:")
    for m in members:
        if m["BorrowedBooks"]:
            print(f"{m['Name']} → Books: {m['BorrowedBooks']}")

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    print()

# -----
# 7. SEARCH Book by Title/Author
# -----
def search_book():
    keyword = input("Enter title or author keyword: ")

    print("\nSearch results:")
    found = False

    for b in books:
        if keyword.lower() in b["Title"].lower() or keyword.lower() in
b["Author"].lower():
            print(f"- {b['Title']} by {b['Author']} (Status:
{b['Availability']})")
            found = True

    if not found:
        print("No matching books found.\n")
    print()

# -----
# 8. Most Popular Genre
# -----
def most_popular_genre():
    if not borrow_log:
        print("No books issued yet.\n")
        return

    genre_count = {}

    for entry in borrow_log:
        g = entry["Genre"]
        if g in genre_count:
            genre_count[g] += 1
        else:
            genre_count[g] = 1

    popular = max(genre_count, key=genre_count.get)

    print(f"\nMost Popular Genre: {popular} ({genre_count[popular]}
issues)\n")

# -----
# MAIN MENU
# -----

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def menu():
    while True:
        print("----- CITY LIBRARY SYSTEM -----")
        print("1. Add Book")
        print("2. Add Member")
        print("3. Issue Book")
        print("4. Return Book")
        print("5. Show Available Books by Genre")
        print("6. Show Members Who Borrowed Books")
        print("7. Search Book")
        print("8. Most Popular Genre")
        print("9. Exit")

        choice = input("Enter your choice: ")

        if choice == "1":
            add_book()
        elif choice == "2":
            add_member()
        elif choice == "3":
            issue_book()
        elif choice == "4":
            return_book()
        elif choice == "5":
            available_books_by_genre()
        elif choice == "6":
            members_with_borrowed_books()
        elif choice == "7":
            search_book()
        elif choice == "8":
            most_popular_genre()
        elif choice == "9":
            print("Exiting system...")
            break
        else:
            print("Invalid choice! Try again.\n")

```

Run the program

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menu()
```

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----- CITY LIBRARY SYSTEM -----
1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book

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- 8. Most Popular Genre
- 9. Exit

Enter your choice: 1
Enter Book ID: 1
Enter Title: Ikigai
Enter Author: Hector Garcia
Enter Genre: Personal Development

Book added successfully!

----- CITY LIBRARY SYSTEM -----

- 1. Add Book
- 2. Add Member
- 3. Issue Book
- 4. Return Book
- 5. Show Available Books by Genre
- 6. Show Members Who Borrowed Books
- 7. Search Book
- 8. Most Popular Genre
- 9. Exit

Enter your choice: 1
Enter Book ID: 2
Enter Title: Emotional Intelligence: Focus
Enter Author: Focal
Enter Genre: Personal Development

Book added successfully!

----- CITY LIBRARY SYSTEM -----

- 1. Add Book
- 2. Add Member
- 3. Issue Book
- 4. Return Book
- 5. Show Available Books by Genre
- 6. Show Members Who Borrowed Books
- 7. Search Book
- 8. Most Popular Genre
- 9. Exit

Enter your choice: 1
Enter Book ID: 3
Enter Title: My fault
Enter Author: john
Enter Genre: Love

Book added successfully!

----- CITY LIBRARY SYSTEM -----

- 1. Add Book

2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 2

Enter Member ID: 1

Enter Name: Vyshu

Enter Age: 22

Enter Contact Info: 9553541695

Member added successfully!

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 2

Enter Member ID: 2

Enter Name: neelima

Enter Age: 24

Enter Contact Info: 9848032919

Member added successfully!

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 3

Enter Member ID: 2

Enter Book ID to issue: 3

Book issued successfully!

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 5

Enter genre: love

Available books in love:

No available books found.

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 4

Enter Member ID: 2

Enter Book ID to return: 3

Book returned successfully!

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 5

Enter genre: love

Available books in love:

- My fault by john

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 6

Members who borrowed books:

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 3

Enter Member ID: 2

Enter Book ID to issue: 2

Book issued successfully!

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 6

Members who borrowed books:
neelima → Books: ['2']

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 7
Enter title or author keyword: ikigai

Search results:
- Ikigai by Hector Garcia (Status: Available)

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 8

Most Popular Genre: Love (1 issues)

----- CITY LIBRARY SYSTEM -----

1. Add Book
2. Add Member
3. Issue Book
4. Return Book
5. Show Available Books by Genre
6. Show Members Who Borrowed Books
7. Search Book
8. Most Popular Genre
9. Exit

Enter your choice: 9

Exiting system...